

Sarvesh Patham

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CAREER OBJECTIVE

Improve industry experience in robotics, imaging/sensor processing, and embedded systems through a **summer internship**

EDUCATION

University of Illinois Urbana Champaign

Urbana, IL

Bachelor of Sciences in Electrical Engineering, Minor in Computer Science

Expected: May 2027

- Cumulative GPA: 3.91
- Relevant Coursework: Analog Signal Processing, Circuit Analysis and Design, Digital Signal Processing, FPGA Lab, MRI Principles/Application

EXPERIENCE

Electrical Engineer Intern

June 2025 – Present

General Autonomy AI

Piscataway, NJ

- Developed LiDAR/Camera/IMU data collection stack using NVIDIA Jetson Orin for simulator development
- Implemented autonomy stack using ROS 2 capable of lanelet planning and object detection from sensor data
- Secured \$50k in investments through LiDAR and Camera mount optimization for investor demos.

Undergraduate Research Assistant

August 2025 – Present

University of Illinois Urbana Champaign

Urbana, IL

- Constructed magnetic field verification system using Python, successfully field tested simulation field lines
- Designed hall effect sensor mount and transceiver + pre-amp housing using Fusion360
- Researching transceiver designs for decreasing signal-to-noise ratio and increasing data clarity for low field MRI

Mechanical Engineering Intern

October 2023 – April 2024

3D Pets

Bloomington, NJ

- Machined/constructed prosthetics to specific consumer standards; increased production rate by 10 percent
- Manufactured body molds and developed 3D Models for prosthetics by scanning molds with LiDAR scanner
- Developed an organizational standard for parts for a newly constructed workshop to efficiently increase workflow

Research Intern

September 2023 – March 2024

New Jersey Institute of Technology

Remote

- Researched electroencephalograms (EEGs) with NJIT Professors Dr. Lipuma and Dr. Bukiet
- Composed an introductory article discussing EEGs and their possible functions in prosthetics
- Presented information at the NJIT STEM for Success conference in front of virtual audience

PROJECTS/EXTRACURRICULARS

Neurotech | Signal Processing Engineer

February 2025 – Present

- Processed EEG data from the frontal lobe/prefrontal cortex by coding digital filters for use in a VR environment
- Filtered data using signal processing techniques including Fast Fourier Transform and wavelet decomposition
- Cleaning noise from signals using filters including notch, bandpass, and high/low frequency filters

Cuff-Link | Embedded Systems Project

December 2023 – May 2024

- Designed a muscle and motion tracking device using Fusion360 and developed in Arduino IDE using C++
- Built a python based virtual cursor controlled by onboard IMU and Electromyography sensor (EMG)
- Selected as a top 3 class finalist for school competition; demonstrated product to audience of 400+
- Gathered data using multiple test subjects to tune cursor for improved accuracy

Illinois Space Society | Hardware Engineer

October 2024 – February 2025

- Developed a board that manages current and voltage monitoring and high-resolution analog to digital converters (ADC) using KiCAD

TECHNICAL SKILLS

Digital: Python, Robotics Operating System(ROS 2), FEM, MATLAB, Linux, C,C++, HTML/CSS, CAD (Fusion360/OnShape), Arduino, ESP32

Hardware: PCB Design(KiCAD), LTspice/SPICE, SystemVerilog, Verilog, Oscilloscopes, Function Generators, Soldering, Communication Protocols (UART/I2C/CAN)